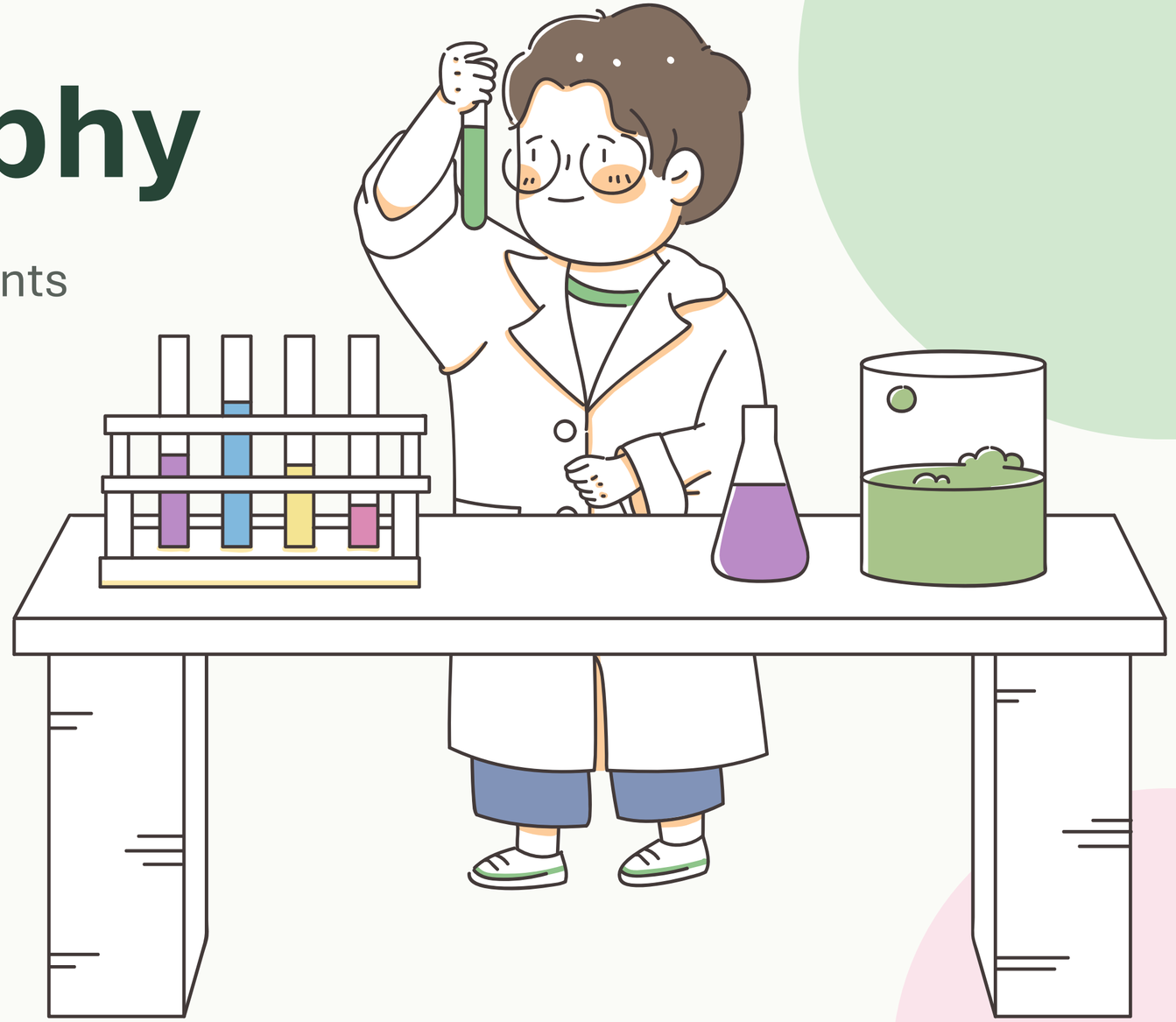
Chromatography

Can we separate the hidden components in a mixture?



INK → PIGMENTS

CHALK → SEPARATE

An Overview

From a sketch pen / ink to a colourful separation.

1. Ask

Is a colour actually just it — or is it hiding many pigments?

2. Explore

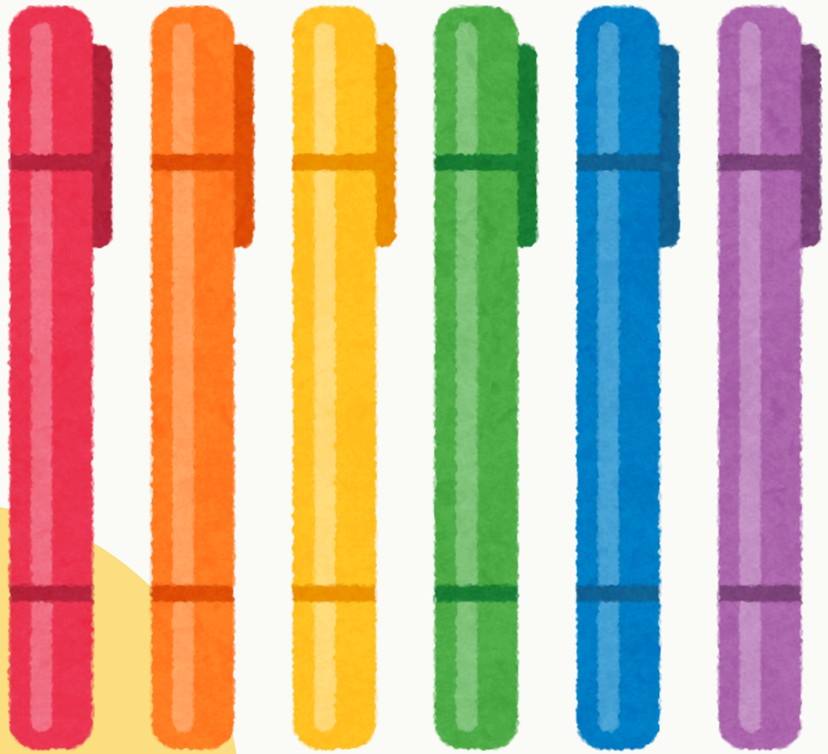
Use a simple chalk strip and solvent to let pigments travel.

3. Observe

Different pigments move different distances, creating visible bands.

Is a Sketch Pen Really that colour?

Look closely: the colour we see can be a mixture of pigments.



NO!
It is a mix of colours.

The challenge

They are mixed together, so they can look like one colour.

Our idea

Make the pigments travel — and see whether they separate.

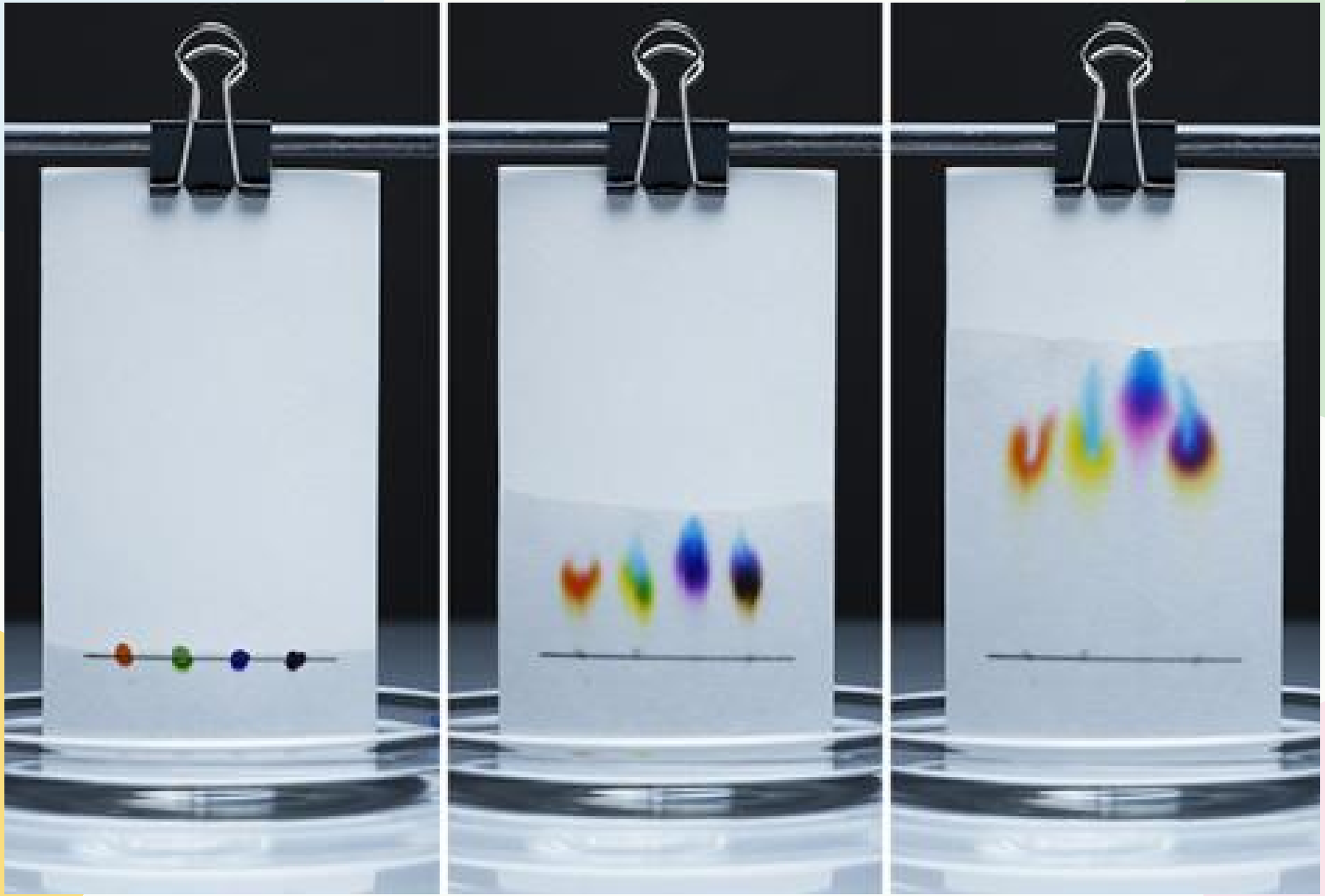
What Is Chromatography?

A simple technique for separating the components of a mixture.

MIXTURE → **SOLVENT MOVES** → **PIGMENTS SEPARATE**



Different substances interact differently with the solid surface and the moving solvent, so they travel at different rates.



Why Are We Using Chalk?

A familiar material can also act as the surface for separation.

Porous structure

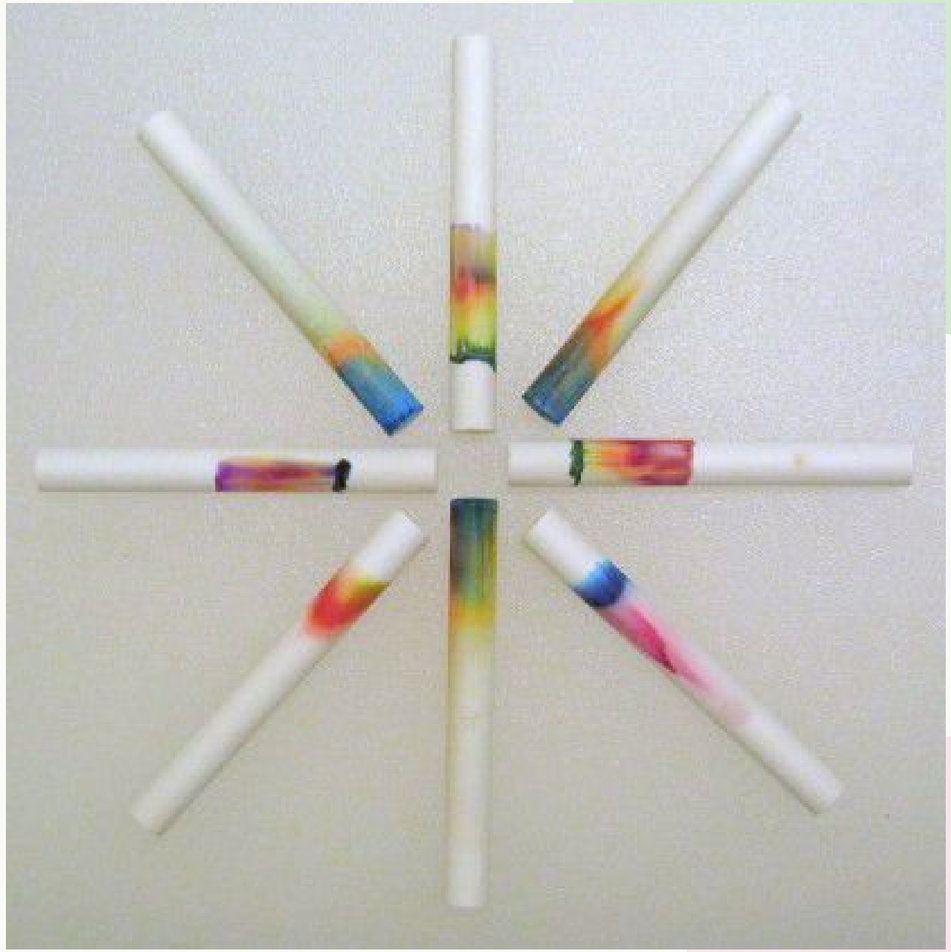
Chalk contains tiny spaces that can allow the solvent to move through it.

Pigment interactions

Different pigments can interact differently with chalk and the solvent.

Outreach advantage

It is simple and easy to do at home also!.



Materials Required

Chalk

A piece of white chalk

Ink / pigment

A washable coloured marker or ink

Solvent

A suitable liquid for the activity

Container

Small cup / beaker

STEP 1: Prepare the Chalk

Make a small, concentrated starting spot.

1

Mark the chalk

Draw a small line or dot of the chosen ink near one end of the chalk.

2

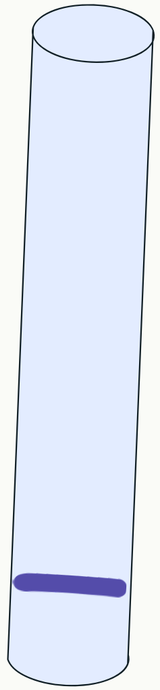
Keep it above the liquid

The ink spot should not be submerged directly in the solvent.

3

Set it upright

Place the chalk so the solvent can move upward through it.



STEP 2: Add the Solvent

The solvent is what carries the pigments.

Add a small amount

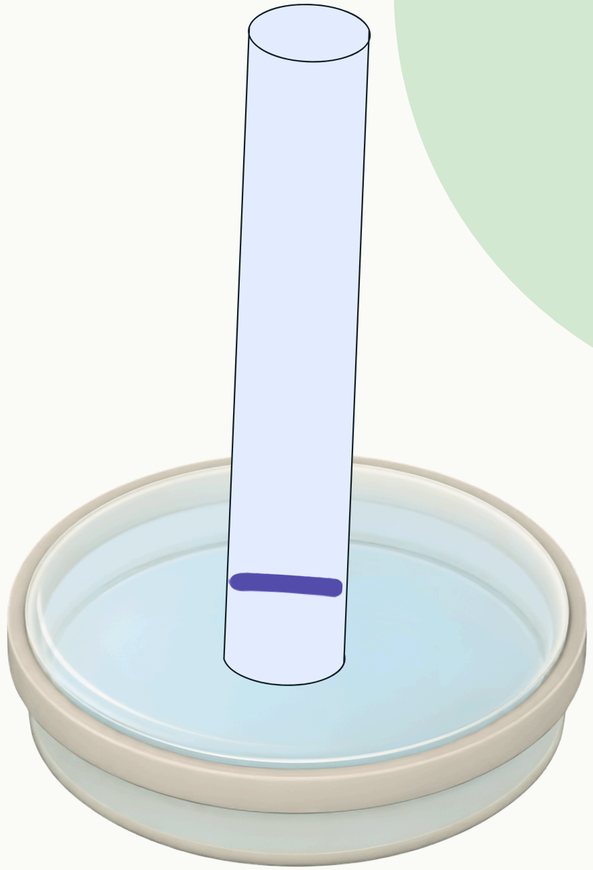
Pour just enough solvent into the container to touch the lower part of the chalk.

Do not cover the ink spot

The starting spot should remain above the solvent level.

Then wait

Let the solvent slowly travel upward.



STEP 3: Let It Travel

Now the separation begins.

The solvent moves through the chalk. Pigments are carried along, but not all at the same speed.

Watch carefully
You may start seeing coloured streaks or bands appear.

Patience matters
Stop when the colours have moved far enough to observe clearly.

STEP 4: Observe the Colours

The final pattern is the evidence of separation.

If one starting colour produces multiple bands, that tells us the original sample contained multiple components.



Chromatography in the Real World

The same idea is used far beyond a classroom experiment.

Biology

Separating and studying pigments, proteins and other molecules.

Medicine

Helping analyse substances in samples and formulations.

Food & environment

Checking mixtures, dyes, additives and contaminants.

One simple experiment introduces a powerful scientific idea: mixtures can be analysed by how their components move.